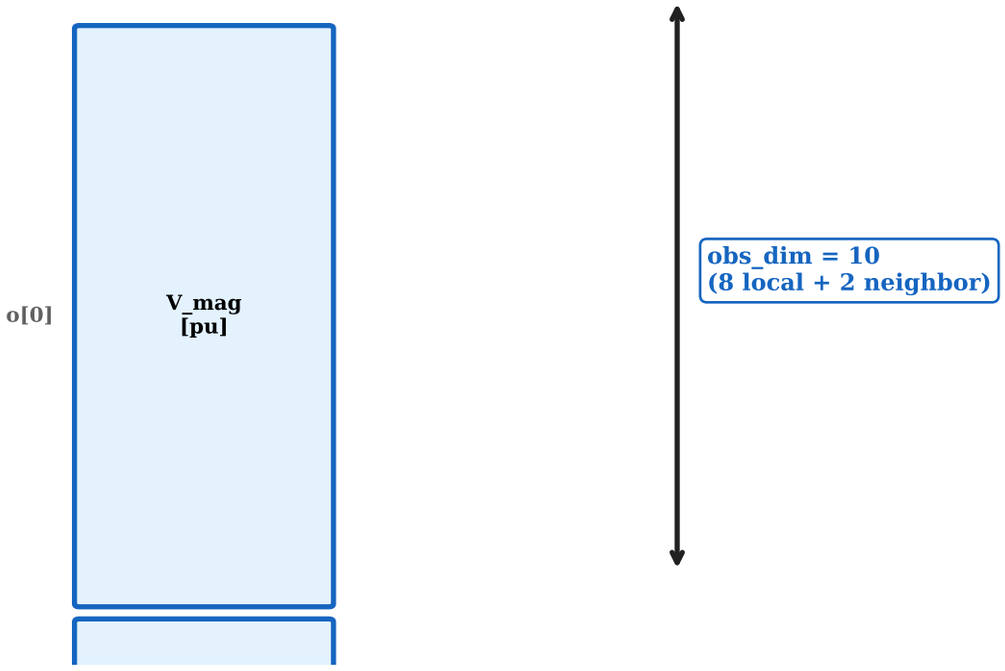
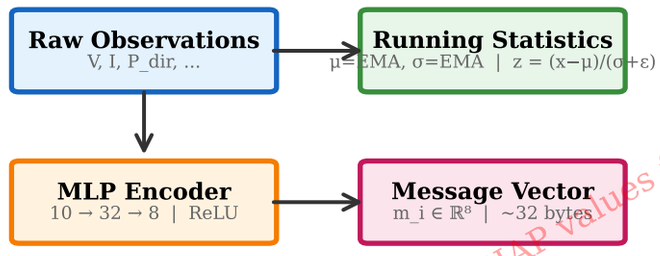
Figure 35: Observation Space Feature Breakdown & Encoding

(a) Observation Vector Structure o_i ∈ ℝ¹⁰



(b) Encoding Pipeline: Raw → Normalized → Embedded



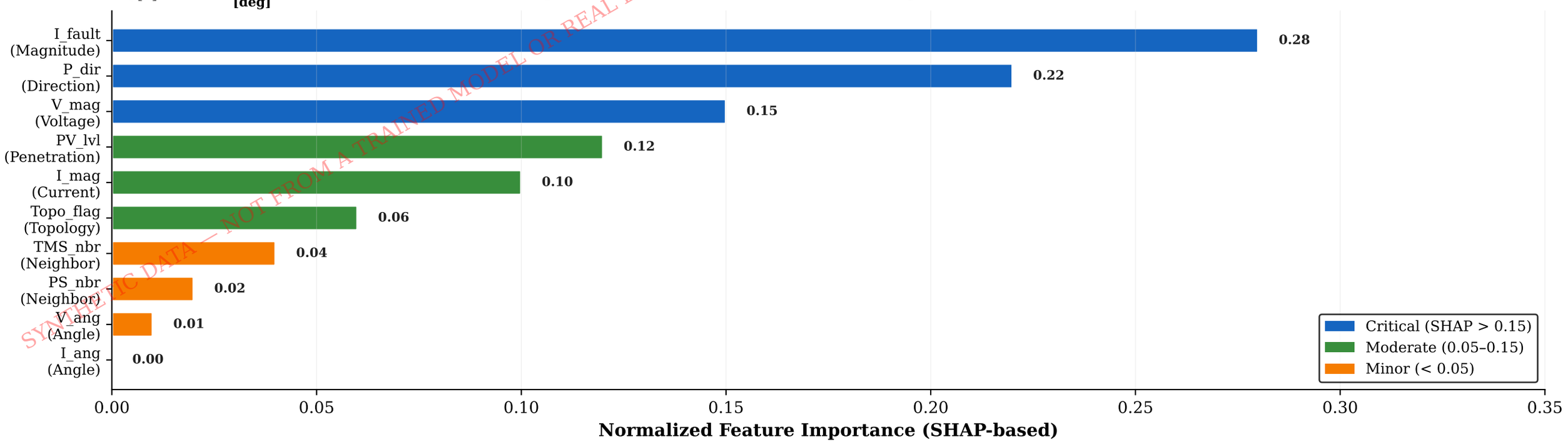
Normalization Details:

- Voltage/Current: Per-unit normalization (base = nominal)
- Power Direction: One-hot encoded {-1, 0, 1}
- Fault Current: Log-scale compression for dynamic range
- PV Level: Normalized to [0, 1] by installed capacity
- Topology Flag: Binary (0 = nominal, 1 = N-1 active)

Message Compression:

- 10-dim raw → 8-float message via learned MLP
- 8 floats × 4 bytes = 32 bytes per message
- At 10 Hz: 32 × 10 × 8 = 2,560 bits/s = 0.256 kbps
- **Well below AMI/RF mesh limit of 1 kbps/agent**

(c) Illustrative Feature Importance for Relay Decision-Making (8-Bus, 100% PV)
[not computed from a trained model — see Figure 48 for real SHAP values]



SYNTHETIC DATA — NOT FROM A TRAINED MODEL OR REAL EXPERIMENT (panel (c) feature-importance bars are illustrative, not SHAP values from a trained model)